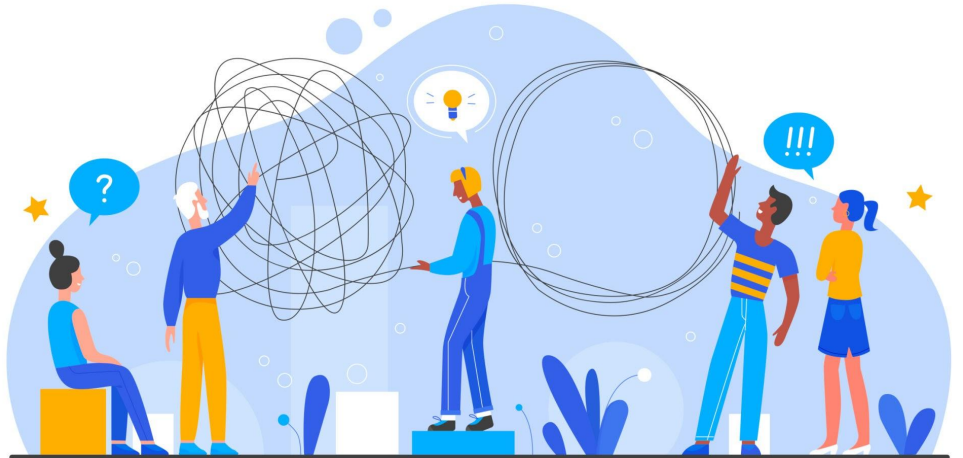


Scaling and Optimizing Enterprise Apps

Architecting for Intelligent Scale

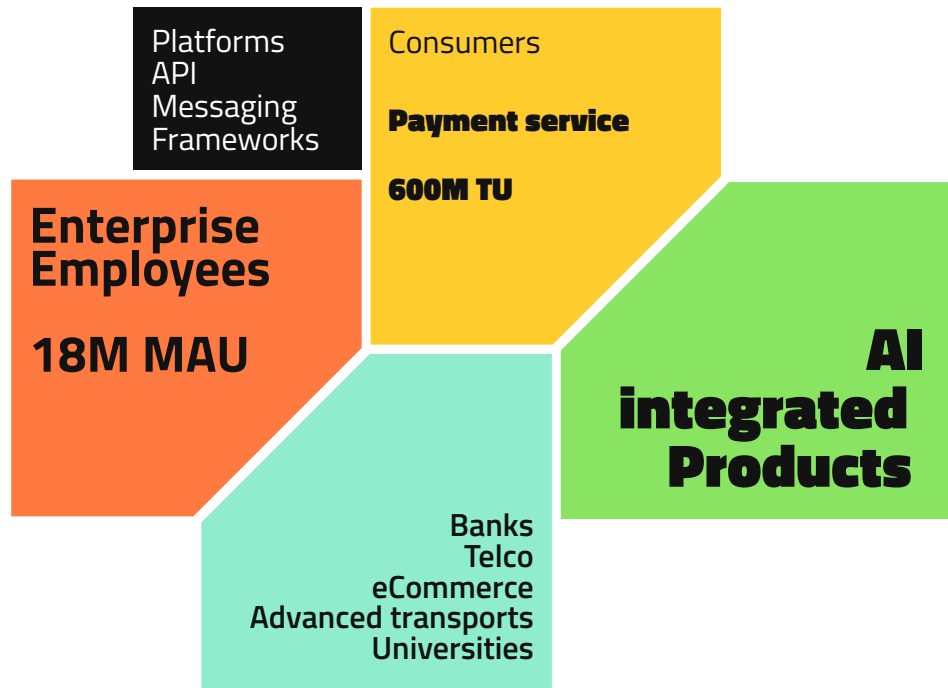
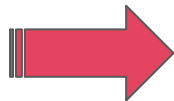
Rohan Rasane

Architect, Servicenow



Learnings / Experience

Sources



The Scaling Illusion

“You can’t solve an architecture problem with more machines.”

Horizontal scaling $\neq$ resilience

True scale = smart decisions, not just infra

Just add nodes/CPU/RAM

Let the cloud deal with it

HA will take care

PMs did not ask me to do that

Architecture is the difference between fifteen users and fifteen million.



What Breaks at Scale

Systems don't fail because of features—they fail because of missing architecture.

Inefficient workflows

Data coupling & retries storms

Fragile async patterns

Invisible tech debt



AI Insight: Use LLMs to surface hidden risks in design/code/traffic patterns

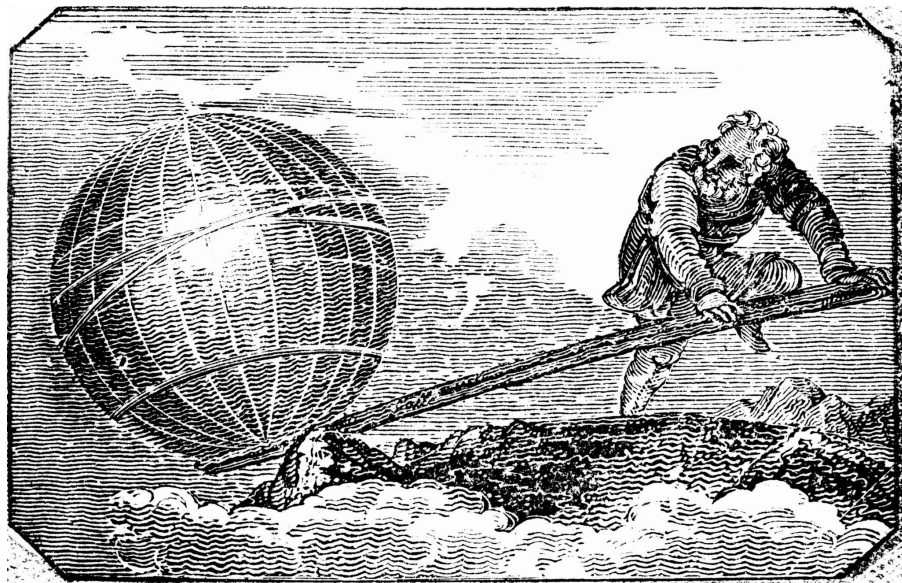
Scale = Architectural Leverage

Smart scale is engineered — not just load-tested.

Isolation & Decoupling

Guardrails

ARB(Architecture Review Board) as
Scale Steward



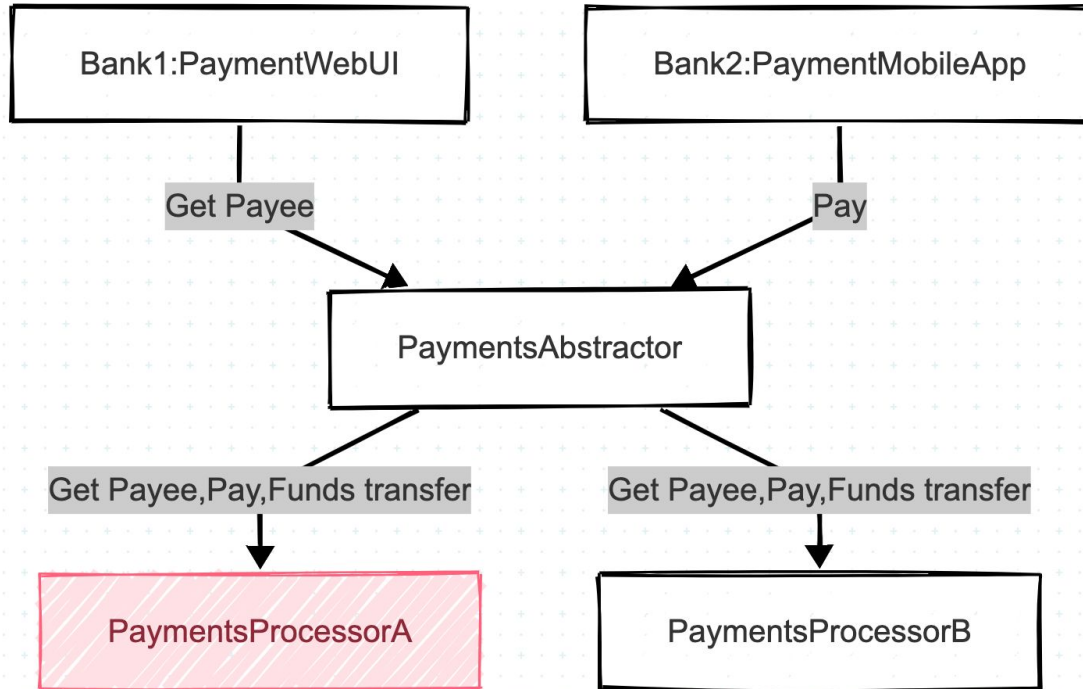
ServiceNow

Founded 2004

Products 1 to 100+

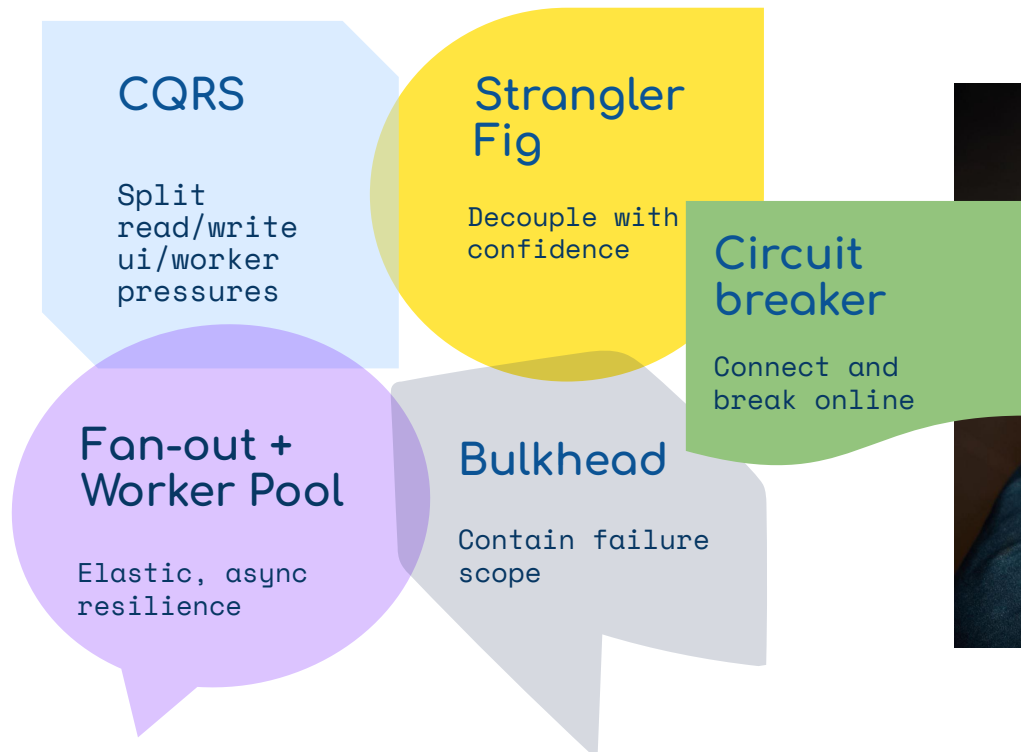
Valuation \$215B

Heyy ! U thereee?



Patterns That Shape Scale

Features ship fast; patterns make them last.



AI Assist: Simulate failure flow, load balance suggestions, dead letter analysis

Frameworks That Enforce Scale

Your architecture is only as repeatable as your defaults.

Integrate Safe patterns in all features: **No Opt outs** **Non-functional requirement**

Pagination, Rate limits, Idempotency

Backpressure, Retry caps

Timeouts + circuit breakers

Guardrails

- Use caching
- Query limit
- Async vs sync guidance



AI Assist: Scan repos for anti-patterns, generate resilience scaffolds

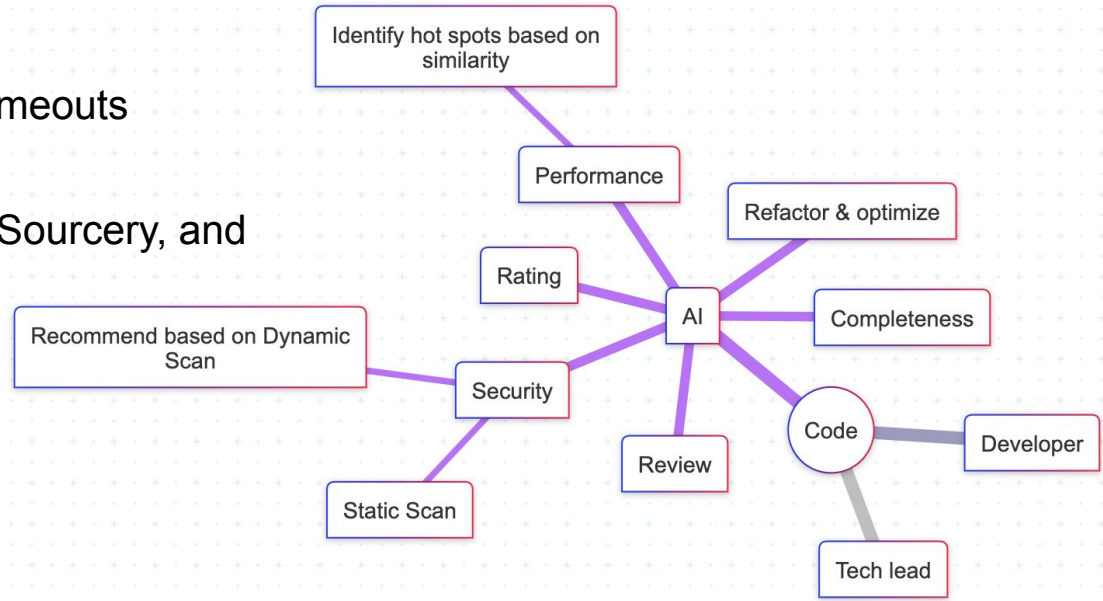
AI as Your Scaling Co-Architect

PR Review: Flag retry loops, missing timeouts

- GitHub : Graphite, Copilot
- Comparable tools : CodeRabbit, Sourcery, and Codium/Qodo

Design Review: Learn and recommend

- ARB library
- Alerts and failures from similar applications

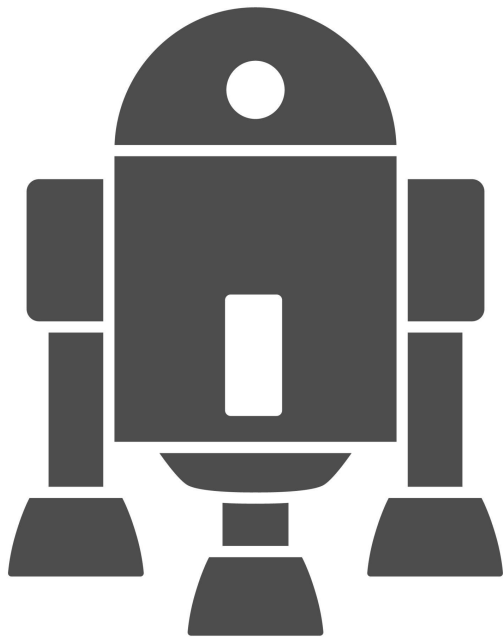


AI as Your Scaling Co-Architect

Failure Event: Correlate logs + metrics for root cause, workflows that detect anomaly trends

- Grep Splunk alerts and app logs
- Cross out False alarms and send alerts

Traffic Surge: Recommend autoscaling trigger adjustments



Think of AI as your second brain, especially for drift detection and design hygiene

ARB as a Scaling Accelerator

Architectural maturity = ability to scale without re-reviewing everything.

Decision records Library: preserve rationale over time

Libraries and criteria help move fast

Enable Shift Left culture



AI-assist: Bucket and recommend patterns, help be more reasonable and intentional

Mindsets to Build Systems That Scale

Architect for scale — don't react to it

Use frameworks to encode best practices

Let AI augment architectural decision-making

Use ARB to scale clarity, not control



Scale intelligently, intentionally, and sustainably

Thank You



<https://github.com/rohanrasane/devsculptcha#readme>

